In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Main)-2017

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST – II
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	C	C
2.	C	B	B
3.	C	C	C
4.	A	D	A
5.	A	C	A
6.	B	B	A
7.	A	B	B
8.	C	C	B
9.	C	A	C
10.	A	A	A
11.	D	B	A
12.	B	C	D
13.	A	D	D
14.	B	A	A
15.	D	B	A
16.	B	B	C
17.	A	A	B
18.	A	A	B
19.	C	B	A
20.	B	B	B
21.	B	A	C
22.	C	C	C
23.	A	C	D
24.	B	B	C
25.	A	C	D
26.	B	A	A
27.	C	C	B
28.	A	B	A
29.	D	B	C
30.	C	A	B

Physics

PART – I

SECTION – A

1. A

Sol. Q – value = Binding energy of reactants – Binding energy of products.

2. C

Sol. $\Delta E = \frac{hc}{\lambda}$ = 59 keV

3. C

Sol. v = 2 cm, u = 3 cm

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \Rightarrow f=\frac{6}{5}cm$

4. A

Sol. $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

On solving,

$\frac{1}{f_1}=\frac{-1}{40}, \frac{1}{f_2}=\frac{1}{60}$

For combination & silvering,

$-\frac{1}{F}=\frac{2}{f_1}+\frac{2}{f_2}-\frac{1}{f_m}$

F = 60 cm

$\frac{1}{v}+\frac{1}{u}=\frac{1}{F}$

On solving with u = –40 cm, F = 60 cm we get V = 24 cm

$v_Z = -\frac{v^2}{u^2}V_o = 1.8\,\hat{i}$

5. A

Sol. Induced emf in the rod $\varepsilon = Blv$

Current in the circuit $I=\frac{\varepsilon}{R}e^{-t/RC}=\frac{Blv}{R}e^{-t/RC}$

Since the net force on the rod should be zero, the external force will be equal in magnitude but opposite to the magnetic force.

$\Rightarrow \quad F = I\,l\,B = \frac{B^2l^2v}{R}e^{-t/RC}$

6. B

Sol. Least count = Main scale / (number of division on the vernier scale) .

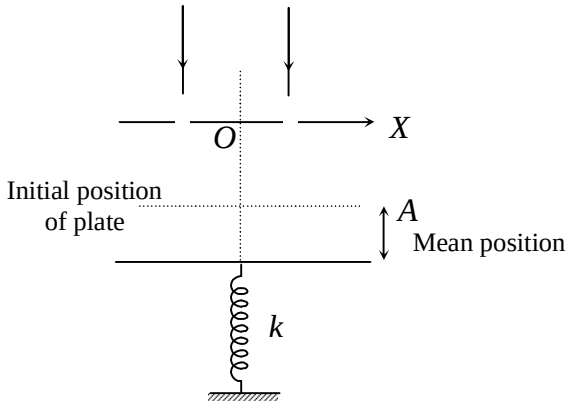
7. A

Sol. $\beta = \dfrac{(D + y)\lambda}{d}$

$\dfrac{d\beta}{dt} = \dfrac{\lambda}{d}\dfrac{dy}{dt} = \dfrac{\lambda}{d}v$

At mean position $v = A\omega = \dfrac{mg}{k}\sqrt{\dfrac{k}{m}} = g\sqrt{\dfrac{m}{k}}$

$\dfrac{d\beta}{dt} = \dfrac{\lambda g}{d}\sqrt{\dfrac{m}{k}}$



8. C

Sol. $F = \dfrac{dP}{dt}$ = rate of change of momentum of the photons. Thus, in case of complete absorption, F = Intensity of the wave / speed of light.

9. C

Sol. $\dfrac{1}{2}2k(2R\theta)^2 + \dfrac{1}{2}I\omega^2 = \text{constant}$

$4kR^2\theta^2 + \dfrac{7}{10}MR^2\omega^2 = \text{constant}$

$4kR^2 2\theta\dfrac{d\theta}{dt} + \dfrac{7}{10}MR^2 2\omega\dfrac{d\omega}{dt} = 0 \quad \Rightarrow \quad T = \pi\sqrt{\dfrac{7M}{10k}}$

10. A

Sol. $\Delta Pm = 2\Delta P_0 \cos kx$ (assuming closed end as origin)

At point *Q*, $x = L - \dfrac{7L}{9} = \dfrac{2L}{9}$

$\Delta Pm = 2\Delta P_0 \cos\left(\dfrac{2\pi}{\lambda} \times \dfrac{2L}{9}\right) = \Delta P_0$

$\therefore$ Required ratio = 1 : 2

11. D

Sol. $E = \dfrac{\rho}{4\varepsilon_o}r$

$V_{AB} = E \times \dfrac{r}{2}$

12. B

Sol. $\frac{E}{4} = 3$

$\Rightarrow$ E = 12

13. A

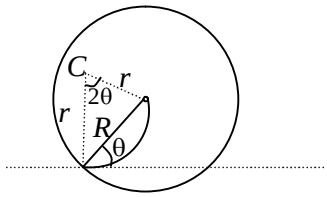
Sol. $E_2 = \frac{E_1}{R_1 + R_{AB}} \times \frac{31.25 \times 10}{50}$

14. B

Sol. $r = \frac{R}{2\sin\theta}$

$\frac{mv_0}{qB} = \frac{R}{2\sin\theta}$

$v_0 = \frac{qBR}{2m\sin\theta}$

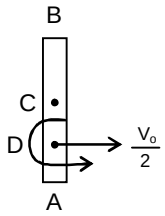


15. D

Sol. By COAM about point D (ie about new centre of mass)

$MV_o \frac{L}{4} = \left[\frac{ML^2}{12} + \frac{ML^2}{16} + \frac{ML^2}{16}\right]\omega$

or, w = $\frac{6v_o}{5L}$



Now, t = $\frac{\theta}{w} = \frac{(\pi/2)}{w} = \frac{5\pi L}{12v_o}$

16. B

Sol. distance = $\int_0^4 |t^2 - 5t + 6|\,dt$

17. A

Sol. $V_o\cos\theta = V\sin\theta$

18. A

Sol. By Bernoulli's theorem: $P + \frac{1}{2}\rho v^2 = \frac{P}{2} + \frac{1}{2}\rho v_1^2 \Rightarrow v_1 = \sqrt{v^2 + \frac{p}{\rho}}$

19. C

Sol. Apply conservation of angular momentum about the point of contact.

20. B

Sol. Rate of decrease of length AB = $\frac{3u}{2}$.

$\therefore \quad AB = d - \frac{3u}{2}t$

Angular velocity = $\frac{v_{tangential}}{AB}$

21. B

Sol. If initial velocities are: $u_1 = \sqrt{2gh}(\downarrow),\ u_2 = \sqrt{2gh}(\uparrow)$

Then final velocities: $v_1 = 3\sqrt{2gh},\ v_2 = \sqrt{2gh}$

By using conservation of momentum and equation of e & m << M.

22. C

Sol. Work done by spring = $U_i - U_f = 3J$

$mx_1 = 2mx_2$ ($\because$ position of COM does not change)

$x_1 = 2x_2$

$\therefore$ Work done by spring on m is $\frac{2}{3}$ of total work.

23. A

Sol. By equation of equilibrium,

$$2T \sin 45° + Mg = \frac{4}{3}\pi R^3 \rho_w g$$

24. B

Sol. Apply Newton's IInd law in y direction and use,

$a_y = \frac{v_y dv_y}{dy}$

25. A

Sol. Apply Newton's IInd law and constraint relations on the blocks.

26. B

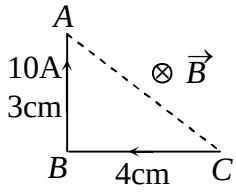
Sol. For equilibrium positions:

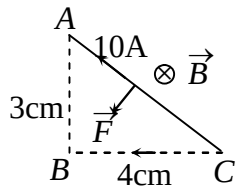
$\frac{dU}{dx} = 0 \quad ; \quad x = 4, 5$

For, x = 5, $\frac{d^2U}{dx^2} = +ve$

27. C

Sol. Force on the conductor ABC
= Force on the conductor AC
= $5 \times 10 \times (5 \times 10^{-2}) = 2.5$ N





28. A
Sol. Factual based

29. D
Sol. $V = 2(x^2 - y^2)$
Equipotential surfaces will be hyperbolic and $\vec{E}$ everywhere will be $\perp$ to them.

30. C

Sol. Potential energy of particle at the centre of earth is $U = -\frac{3}{2}\frac{GMm}{R_e}$

So, $V_e = \sqrt{\frac{3GM}{R_e}} = \sqrt{3gR_e}$

Chemistry

PART – II

SECTION – A

1. C
Sol. $P = K_2SO_4$, $Q = BaSO_4$, $R = BaS$, $S = H_2S$

2. B
Sol. $A = H_2B_4O_7$, $B = H_3BO_3$, $C = B_2O_3$, $D = MgO$
$E = Mg(NO_3)_2$, $F = NO_2$
$D(MgO) + H_2O \longrightarrow Mg(OH)_2$

3. C
Sol. The acid (C) is most acidic due to presence of least number of +I groups in it.

4. D
Sol. $C_2H_5OH + Na \longrightarrow C_2H_5ONa + H(\text{nascent hydrogen})$
$H + H \longrightarrow N_2$

5. C
Sol. London smog is caused by SO_2 and Los Angeles smog is caused by NO_2

6. B
Sol. Na_2CO_3 decomposes above 1100^oC

7. B
Sol. $Cl_2 + FeCl_3 \longrightarrow FeCl_4^- + Cl^+$
The electrophile Cl^+ substitutes a hydrogen atom from benzene.

8. C
Sol. Phosphates accelerate the growth of plants in water, which consume the dissolved oxygen

9. A
Sol. It is produced in marshy places unlike other hydrocarbons.

10. A
Sol. $Be \longrightarrow 1s^2 2s^2 2p^0$
$Be^{2+} \longrightarrow 2s^0 2p^0$ (Four vacant orbitals)
$\therefore$ Be^{2+} forms complex with four water molecules.

11. B
Sol. H_2O_2 forms more number of intermolecular hydrogen bond than water.

12. C

Sol. The ions that are responsible for hardness of water are Ca^{2+}, Mg^{2+}, HCO_3^-, Cl^- and SO_4^{2-}.

13. D

Sol. At normal boiling point (500 K), the following equilibrium exists between the two physical states at 1 atm.

$$X(\text{liquid}) \rightleftharpoons X(\text{gas})$$

For the equilibrium $\Delta G = 0$

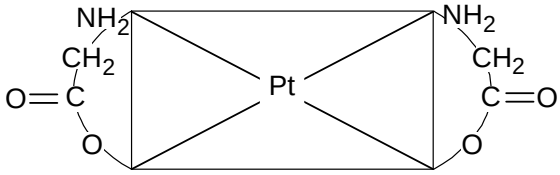
If the pressure increases the change shifts to backward direction for which ΔG = +ve.

If pressure decreases the change shifts towards forward direction, for which ΔG = -ve.

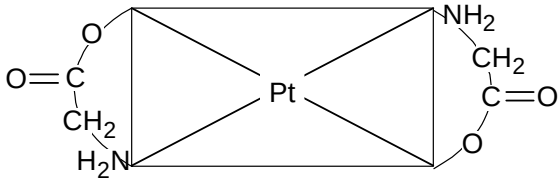
If temperature increase, the change will proceed towards forward direction, for which ΔG = -ve.

14. A

Sol. The geometrical isomers of (B) are

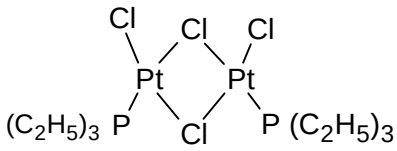


Cis

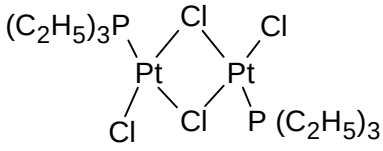


trans

The geometrical isomers of (C) are

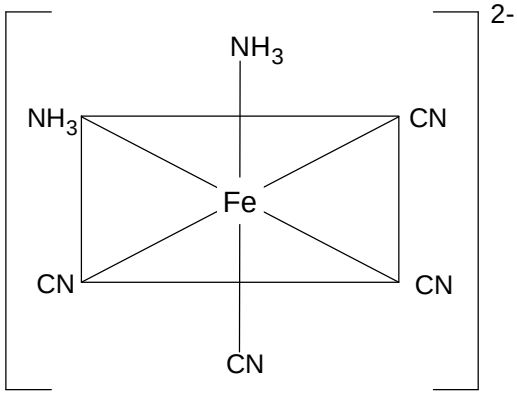


Cis

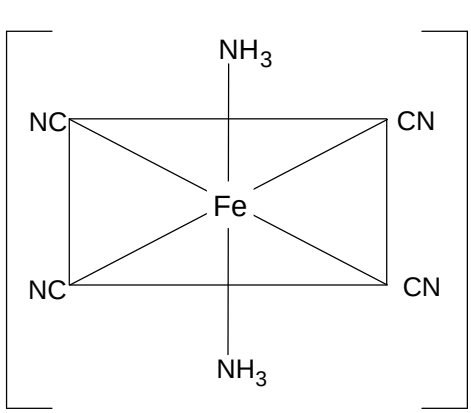


trans

The geometrical isomers of (D) are



Cis



trans

15. B

Sol. $\Delta C_p = 2 \times 25.1 + 3 \times 75.3 - (103.8 + 3 \times 28.8) = 85.9 \text{ J/mole}$

$$\frac{\Delta H_{358} - \Delta H_{298}}{T_2 - T_1} = \Delta C_p$$

$\Delta H_{358} = -28.136 \text{ kJ/mole}$

16. B

Sol. In case of weak acid heat of neutralization is greater than – 13.7 KCl / mole because extra energy is required to dissociate the acid

17. A

Sol. pH = 11 $\Leftrightarrow$ pOH = 14–11 = 3, $[OH^-] = 10^{-3}$ M

$10^{-3} = \frac{n_{NaOH}}{1} \Rightarrow$ mass of NaOH = 40×10^{-3} gm = 0.04 gm

18. A

Sol. ${}_{90}Th^{228} \longrightarrow {}_{83}Bi^{212}$ No. of α-particle = $\frac{228-212}{4} = 4$

No. of β-particle = 1

19. B

Sol. $NaX \longrightarrow Na^+ + X^-$

$$\underset{t=0}{} \quad X^- + H_2O \rightleftharpoons HX + OH^-$$

$$t=0 \quad 0.1.0 \quad \text{excess} \quad 0 \quad 0$$

$$t=\text{eq.} \quad 0.1(1-\alpha) \quad 0.1\alpha \quad 0.1\alpha$$

$$K_h = \frac{[HX]\,[OH^-]}{[X^-]}$$

$$= \frac{[HX]}{[H^+]\,[X^-]} \times \frac{[H^+][OH^-]}{1}$$

$$= \frac{K_w}{K_a} = \frac{0.1\alpha \times 0.1\alpha}{0.1(1-\alpha)} \simeq 0.1\,\alpha^2 \quad [1-\alpha \approx 1 \text{ as } 1 \gg \alpha \text{ for weak acid}]$$

$$\therefore \frac{10^{-14}}{10^{-5}} = 0.1\,\alpha^2$$

$$\therefore \alpha = 1\times10^{-4}$$

% hydrolysis = 0.01%

20. B

Sol. pOH = pKb + log $\frac{(NH_4^+)}{(NH_3)} = pK_b$

$(OH^-) = 1.8\times10^{-5}$

Ksp=$[Mn^{2+}]\,[OH^-]^2$, $[Mn^{2+}] = 1.38\times10^{-4}$

21. A

Sol. (A) As we know that,

Resonance energy = $\Delta H°$ (observed) – $\Delta H°$ (calculated)

= (50.4 – 232.6) kJ / mol = – 182.2 kJ mol^{-1}

22. C

Sol. $t_{1/2}$ is independent of concentratin of reactant for 1st order reaction

23. C

Sol. Increases in pressure or addition of C affect the second equilibrium hence concentration of A_2 changes which further affect the first equilibrium. Addition of inert gas at constant volume does not affect the equilibrium

24. B

Sol. $HIn \rightleftharpoons H^+ + In^-$

$$K_a = \frac{\left[H^+\right]\left[In^-\right]}{\left[HIn\right]} = \frac{\left[H^+\right][base]}{[Acid]}$$

$$\text{or, } \left[H^+\right] = Ka\frac{[Acid]}{[Base]}$$

For 75% red, $\left[H^+\right] = \frac{Ka \times 75}{25} = \frac{3\times10^{-5}\times75}{25} = 9\times10^{-5}$; $pH = 4.05$

For 75% blue, $\left[H^+\right] = \frac{\left(3\times10^{-5}\right)\times25}{75} = 1\times10^{-5}$; $pH = 5$

The change in pH = 5-4.05 = 0.95

25. C

Sol. $\frac{r_1}{r_2} = \frac{K[A_1]^a[B_1]^b}{K[A_2]^a[B_2]^b}$ Apply

26. A

Sol. $Na_2B_4O_7.10H_2O \xrightarrow[-10H_2O]{heat} Na_2B_4O_7 \xrightarrow{740^oC} \underset{\text{Sodium metaborate}}{2NaBO_2} + B_2O_3$

$B_2O_3 + CuSO_4 \longrightarrow CuO.B_2O_3 + SO_3$

$\downarrow$

$Cu(BO_2)_2$ (blue)

Copper metaborate

27. C

Sol. $H_3BO_3 + 3C_2H_5OH \longrightarrow \underset{\text{(green colour)}}{B(OC_2H_5)} + 3H_2O$

28. B

Sol. $AlF_3 + 3KF \longrightarrow K_3\left[AlF_6\right]$

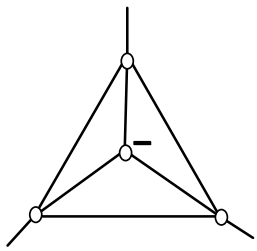
For this reaction, maximum concentration of F^- ions is obtained from KF.

29. B

Sol. $S_1 = \frac{K_{sp}}{0.25}S_2^2 = \frac{K_{sp}}{4\times0.5} = \frac{K_{sp}}{2}$

30. A

Sol.



Mathematics

PART – III

SECTION – A

1. C

Sol. From the figure, $12 = x + r + 8$

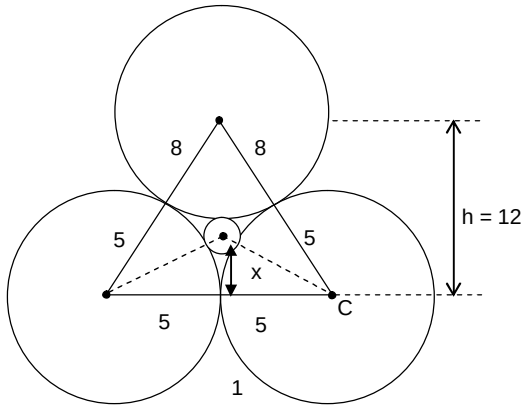
$\Rightarrow 12 = \sqrt{(r+5)^2 - 25} + r + 8$

$\Rightarrow 12 = \sqrt{r^2 + 10r} + r + 8$

$\Rightarrow (4 - r)^2 = r^2 + 10r$

$\Rightarrow 18r = 16$

$\Rightarrow r = \frac{8}{9}$



2. B

Sol. $x^2\left(3 + 2^{\sqrt{\log_2 3}} - 3^{\sqrt{\log_3 2}}\right)x - 2\left(3^{\log_3 2} - 2^{\log_2 3}\right) = 0$

$\Rightarrow x^2 - 3(x) - 2(2-3) = 0$

$\Rightarrow x^2 - 3x + 2 = 0$

$\Rightarrow \alpha = 2, \beta = 1$

$\therefore \quad \alpha^2 + \alpha\beta + \beta^2 = 4 + 2 + 1 = 7$

3. C

Sol. $f'(x) = e^x + 3$

$f'(x) > 3$

$\frac{f(x+3) - f(x-5)}{8} = f'(c) > 3$

$\Rightarrow f(x+3) - f(x-5) > 24$

4. A

Sol. We must have $e_1 < 1 < e_2$

$\Rightarrow f(1) < 0 \Rightarrow a > 3$

5. A

Sol. $\theta_1 + \theta_2 = \frac{\pi}{2} + \frac{\pi}{2} = \pi$

$\cos\theta_2 = \left(\frac{4}{5}\right)\left(\frac{1}{3}\right) - \left(\frac{3}{5}\right)\left(\frac{\sqrt{8}}{3}\right) = \frac{4 - 6\sqrt{2}}{15}$,

a negative number

Therefore, θ_2 is obtuse and hence θ_1 is acute.

$\therefore \quad \theta_1 < \theta_2$

6. A

Sol. Taking P and Q as t_1 and t_2, respectively

The equation to PQ is $-2x + y(t_1 + t_2) = 2t_1t_2$

Comparing with $lx + my = 1$, we get $t_1 + t_2 = -\frac{2m}{l}$ and $t_1t_2 = -\frac{1}{l}$

R is $\left(\frac{1}{t_1^2} - \frac{2}{t_1}\right)$ and T is $\left(\frac{1}{t_2^2} - \frac{2}{t_1}\right)$

[$\because$ PSR and QST are focal chords]

Slope of $RT = \frac{2\left(\frac{1}{t_2} - \frac{1}{t_1}\right)}{\frac{1}{t_1^2} - \frac{1}{t_2^2}}$

$= -\frac{2t_1t_2}{t_1 + t_2} = \frac{-2\left(-\frac{1}{l}\right)}{-\frac{2m}{l}} = -\frac{1}{m}$

7. B

Sol. $y = \frac{\sqrt{3}x}{2}$

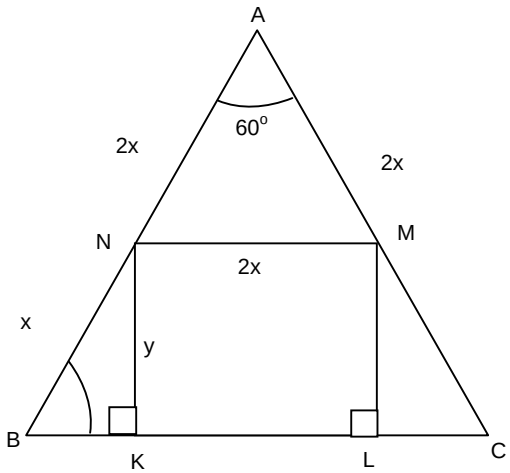
$z = \frac{x}{2}$

$\frac{1}{2}yz = 6 \Rightarrow x^2 = \frac{48}{\sqrt{3}}$

Area of $\Delta ABC = 6 + 6 + 2xy + \frac{1}{2}(2x)(2x)\sin 60^o$

$= 12 + 2x\frac{\sqrt{3}x}{2} + 2x^2\frac{\sqrt{3}}{2}$

$12 + 2\sqrt{3} \times \frac{48}{\sqrt{3}} = 108$



8. B

Sol. $\tan A + \tan B + \tan C = \lambda$

$\Rightarrow \tan A\left(\tan^2 B\right) = \lambda\left(B = C\right)$

$\Rightarrow \tan^2 B = \left(\dfrac{\lambda}{\tan A}\right)$, $A > 90^o$, has infinitely many solutions.

9. C

Sol. $I = \int_0^{\pi/2} \sin x \sin 2x \sin 3x \sin 4x\, dx$ .....(i)

Replacing x by $\dfrac{\pi}{2} - x$, we get

$I = \int_0^{\pi/2} \cos x \sin 2x \cos 3x \sin 4x\, dx$ .....(ii)

Adding (i) and (ii), we get

$2I = \int_0^{\pi/2} \cos 2x \sin 2x \sin 4x dx = \dfrac{\pi}{8}$

$\therefore I = \dfrac{\pi}{16}$

10. A

Sol. $\sqrt{x+y} + \sqrt{y-x} = c$ .....(i)

$\Rightarrow \quad \dfrac{\left(y+x\right)-\left(y-x\right)}{\sqrt{x+y}-\sqrt{y-x}} = c$

$\Rightarrow \quad \sqrt{y+x} - \sqrt{y-x} = \dfrac{2x}{c}$ .....(ii)

By adding (i) and (ii), we get $2\sqrt{y+x} = c + \dfrac{2x}{c}$

$\Rightarrow \quad 4\left(y+x\right) = c^2 + \dfrac{4x^2}{c^2} + 4x$

$\therefore \quad 4\dfrac{dy}{dx} = \dfrac{8x}{c^2}$

$\therefore \quad \dfrac{d^2y}{dx^2} = \dfrac{2}{c^2}$

11. A

Sol. A and B are $\left(\sqrt{5}, 1\right)$ and $\left(-\sqrt{5}, -1\right)$.

Let C be $\left(2\sec\theta, 2\tan\theta\right)$.

$O\left(0,0\right)$ is the mid point of AB.

Slope of $OC = \sin\theta$ and slope of $AB = \dfrac{1}{\sqrt{5}}$

Since $OC \perp AB$,

$\sin\theta = -\sqrt{5}$ which is impossible

12. D

Sol. $\frac{k}{1}=\frac{\sin\left(A+\frac{C}{2}\right)}{\sin\frac{C}{2}}$

$\Rightarrow \frac{k+1}{k-1}=\frac{\sin\left(A+\frac{C}{2}\right)+\sin\frac{C}{2}}{\sin\left(A+\frac{C}{2}\right)-\sin\frac{C}{2}}=\cot\frac{B}{2}\cot\frac{A}{2}$

13. D

Sol. Number of diagonals passing through centre = 6

Number of rectangles $={}^6C_2=15$

14. A

Sol. Putting $z=x+iy,$ we get

$(x+iy)^2+(p+iq)(x+iy)+r+is=0$

$\Rightarrow (x^2-y^2+px-qy+r)+i(2xy+py+qx+s)=0$

$\Rightarrow x^2-y^2+px-qy+r=0$ ……(i)

and $2xy+py+qx+s=0$ ……(ii)

If the roots are real, then $y=0$

$\therefore$ (i) gives $x^2+px+r=0$ ……(iii)

and (ii) gives $qx+s=0$ ……(iv)

From (iv), $x=-\frac{s}{q}$

Putting in (iii), we get $\frac{s^2}{q^2}-\frac{ps}{q}+r=0$

$\Rightarrow \quad pqs=s^2+rq^2$

15. A

Sol. $y=2\log_a x$

$\Rightarrow \log_a x=\frac{y}{2}$

$\Rightarrow x=a^{y/2}$

$\Rightarrow f^{-1}(y)=a^{y/2}$

$\Rightarrow f^{-1}(b+c)=a^{\frac{b+c}{2}}=f^{-1}(b).f^{-1}(c)$

16. C

Sol. $f(x)=\frac{1}{1+\frac{1}{x}}=\frac{x}{x+1}$ is not differentiable at x = 0, –1.

Also, $g(x)=\dfrac{1}{1+\dfrac{2}{f(x)}}=\dfrac{1}{1+\dfrac{2(x+1)}{x}}=\dfrac{x}{3x+2}$

Thus, the point where $g(x)$ is not differentiable are $x=0,-1,-2/3$

17. B

Sol. Point of intersection of curves is (1, 0)

$y=3^{x-1}\ln x$

$\Rightarrow y'=\dfrac{3^{x-1}}{x}+\ln x.3^{x-1}.\ln 3$

$\therefore m_1=1$

Now, $y=x^x(1+\ln x)$

$\therefore\ m_2=1$

So, $\tan\theta=\tan\theta=\left|\dfrac{1-1}{1+1}\right|=0$

$\Rightarrow\theta=0\Rightarrow\cos 0^o=1$

18. B

Sol. $B=A_1+3A_3^3+....+(2n-1)(A_{2n-1})^{2n-1}$

$B^T=-\left[A_1+3A_3^3+.....+(2n-1)(A_{2r-1})^{2r-1}\right]$

$=-B$

Thus, B is skew symmetric.

19. A

Sol. The possible arrangements of teacher and students can be as follows:

(i) T S S T S S T S S

(ii) S T S S T S S T S

(iii) S S T S S T S S T

Hence, total number of ways $=3(3!)6!=(18)6!$

20. B

Sol. $y-x=3\left(\dfrac{1}{2^2}+\dfrac{1}{4^2}+\dfrac{1}{6^2}+.....\right)$

$x-z=\left(\dfrac{1}{2^2}+\dfrac{1}{4^2}+\dfrac{1}{6^2}+.....\right)$

$\therefore (y-x)=3(x-z)$

$4x=y+3z$

$\Rightarrow\ \Rightarrow 2\dfrac{x}{3}=\dfrac{y}{6}+\dfrac{z}{2}$

21. C

Sol. Let O = (0, 0), P = (6, 8) and Q = (12, 16).

As shown in the figure the shortest route consists of tangents OT, minor arc TR and tangent RQ.

Since $OP = 10,\ PT = 5,$ and $\angle OTP = 90^o$

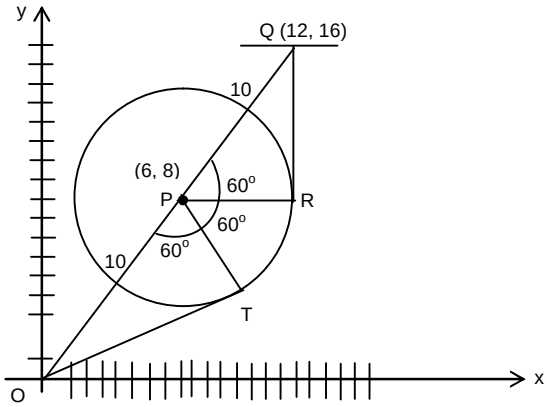
It follows that $\angle OPT = 60^o$ and $OT = 5\sqrt{3}$.

Similarly, $PQ = 10,\ PR = 5$

$\therefore\ \angle QPR = 60^o$ and $QR = 5\sqrt{3}$

Also points O, P and Q are collinear as $m_{OP} = m_{PQ}$

$\therefore \angle RPT = 60^o,$ so arc TR is of length $\dfrac{5\pi}{3}$.



22. C

Sol. Given $\int_0^4 f(x)\,dx - \int_0^4 g(x)\,dx = 10$

$\Rightarrow (A_1 + A_3 + A_4) - (A_2 + A_3 + A_4) = 10$

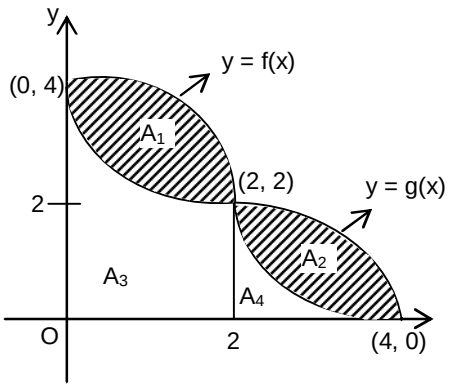
$\Rightarrow\ A_1 - A_2 = 10$ (i)

Also $\int_2^4 g(x)\,dx - \int_2^4 f(x)\,dx = 5$

$\Rightarrow (A_2 + A_4) - A_4 = 5$

$\Rightarrow\ A_2 = 5$ (ii)

Adding (i) and (ii), $A_1 = 15$



23. D

Sol. $(ABA)^2 = (ABA)(ABA) = AB(AA)BA = AB^2A$

$(ABA)^3 = (AB^2A)(ABA)(ABA) = (AB^3A)$

$(ABA)^4 = (AB^3A)(ABA) = AB^4A$

Continuing in this manner, we get $C^{2009} = AB^{2009}A$

$$B^2 = \begin{bmatrix} 1 & \sqrt{2} \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & \sqrt{2} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2\sqrt{2} \\ 0 & 1 \end{bmatrix}$$

$$B^3 = \begin{bmatrix} 1 & 2\sqrt{2} \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & \sqrt{2} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3\sqrt{2} \\ 0 & 1 \end{bmatrix}$$

Continuing in this manner, we get $B^{2009} = \begin{bmatrix} 1 & 2009\sqrt{2} \\ 0 & 1 \end{bmatrix}$

$\therefore \quad C^{2009} = A\begin{bmatrix} 1 & 2009\sqrt{2} \\ 0 & 1 \end{bmatrix}A$

24. C

Sol. $\lim\limits_{x\to\infty} x^2 \sin\left(\log_e \sqrt{\cos\frac{\pi}{x}}\right)$ (Form $0\times\infty$)

$$= \lim_{x\to\infty} \frac{x^2 \sin\left(\log_e \sqrt{\cos\frac{\pi}{x}}\right)}{\left(\log_e \sqrt{\cos\frac{\pi}{x}}\right)}\left(\log_e \sqrt{\cos\frac{\pi}{x}}\right)$$

$$= \lim_{x\to\infty} x^2.\log_e \sqrt{\cos\frac{\pi}{x}} = \lim_{x\to\infty} \frac{x^2}{2}\log_e\left(\cos\frac{\pi}{x}\right)$$

$$= \lim_{x\to\infty} \frac{x^2 \log_e\left[1+\left(\cos\frac{\pi}{x}-1\right)\right]}{2\left(\cos\frac{\pi}{x}-1\right)}\left(-2\sin^2\frac{\pi}{2x}\right)$$

$$= \lim_{x\to\infty} \frac{x^2}{2}\frac{\left(-2\sin^2\frac{\pi}{2x}\right)}{\left(\frac{\pi}{2x}\right)^2}.\left(\frac{\pi}{2x}\right)^2 = -\frac{\pi^2}{4}$$

25. D

Sol. $I_1 = \int_0^\infty \frac{x^2\sqrt{x}}{(1+x)^6}dx$

Let $x = \frac{1}{t} \quad \Rightarrow I_1 = \int_\infty^0 \frac{\frac{1}{t^2\sqrt{t}}}{\left(1+\frac{1}{t}\right)^6}\left(-\frac{1}{t^2}dt\right)$

$\Rightarrow \; I_1 = \int_0^\infty \frac{t\sqrt{t}}{(1+t)^6}dt = I_2$

26. A

Sol. The equation of AB is x – y = 1.
Let the orthocenter be (h, k)

Its mage w.r.t. AB is $(k+1, h-1)$,

which will lie on the circle, so $(k+1)^2 + (h-1)^2 = 5$

27. B

Sol. For $x \in \left(0, \frac{\pi}{2}\right)$

$L.H.S. = \cos x(\cos x - 1) < 0$

$R.H.S = x$, positive as $x \in \left(0, \frac{\pi}{2}\right)$

$\therefore$ L.H.S. $\neq$ R.H.S

So, $x = 0$ is only solution

28. A

Sol. $\sum_{r=1}^{n} \tan^{-1}\left(\frac{2r-1}{(r^2+r+1)(r^2-r+1)-2r^3}\right)$

$= \sum_{r=1}^{n} \tan^{-1}\left(\frac{2r-1}{r^4+r^2+1-2r^3}\right) = \sum_{r=1}^{n} \tan^{-1}\left(\frac{2r-1}{1+r^2(r^2-2r+1)}\right)$

$= \sum_{r=1}^{n} \tan^{-1}\left(\frac{r^2-(r-1)^2}{1+r^2(r-1)^2}\right) = \sum_{r=1}^{n} \left(\tan^{-1}(r)^2 - \tan^{-1}(r-1)^2\right)$

$= \tan^{-1}(n^2)$

$\therefore \tan\left(\tan^{-1}(n^2)\right) = 961$

$\Rightarrow n^2 = 961 \Rightarrow n = 31$

29. C

Sol. $2\left({}^7C_1 + {}^7C_2 + {}^7C_3\right)$

$= 2(7 + 21 + 35) = 126$

30. B

Sol. $P\left(\frac{F}{G}\right) = \frac{P(F \cap G)}{P(G)} = \frac{4}{10} \cdot \frac{5}{4} = \frac{1}{2}$

